

VoiceNavigator

Scalable Voice-Driven Mobile App for
Regulatory Compliance and ESG Impact

CMPE 277 - Smartphone Application Development
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1. Executive Summary

VoiceNavigator is a next-generation, touchless Android mobile application powered by voice-first navigation. Built with Kotlin and Jetpack Compose, it delivers two high-impact solutions: (1) an ESG Market Research Navigator with real-time data visualization from World Bank and BLS APIs, and (2) a California DMV Knowledge Test Preparation platform using RAG (Retrieval Augmented Generation) architecture.

The app targets enterprise compliance needs for a consulting firm whose tours, travel, and on-demand charter services require employees to stay current with evolving DMV regulations, while also providing ESG and food security analytics capabilities.

Key achievements include a fully functional voice navigation system, real-time API data visualization with custom Canvas-based charts, a complete RAG pipeline with both neural and TF-IDF embeddings, and robust offline fallback for all features.

2. Technology Stack

Component	Technology
Language	Kotlin
UI Framework	Jetpack Compose + Material 3
Architecture	MVVM (Model-View-ViewModel)
Networking	OkHttp + Retrofit + HttpURLConnection
JSON Parsing	Gson + org.json
Async	Kotlin Coroutines + StateFlow
Charts	Custom Canvas-based (Compose)
Voice	Android SpeechRecognizer + TextToSpeech
AI / LLM	Google Gemini 2.0 Flash
Embeddings	Gemini text-embedding-004 + TF-IDF fallback
PDF Extraction	PdfBox Android 2.0.27
Min SDK	26 (Android 8.0)
Target SDK	34 (Android 14)

3. System Architecture

3.1 MVVM Architecture

The app follows the Model-View-ViewModel pattern with clear separation of concerns:

- **Presentation Layer:** Jetpack Compose screens (MarketResearchScreen, ESGDashboardScreen, DMVScreen) with reusable components (VoiceButton, Charts, BottomNavBar).
- **ViewModel Layer:** MarketResearchViewModel, ESGViewModel, DMVViewModel manage UI state via Kotlin StateFlow, handle API calls, caching, and fallback logic.

- **Data Layer:** API services (GeminiApiService, WorldBankApiService, BLSApiService) and the RAGEngine handle all external data fetching and processing.
- **Voice Layer:** VoiceRecognitionManager handles STT/TTS, VoiceCommandParser maps speech to sealed-class commands routed through MainActivity.

3.2 RAG (Retrieval Augmented Generation) Pipeline

The RAGEngine implements a full vector-based retrieval pipeline matching the professor's architecture:

- **Data Sources:** SOFI 2024/2025 food security reports and CA Driver's Handbook content bundled in-app.
- **Text Chunking:** ~500-word chunks with 50-word overlap for context continuity.
- **Embedding Generation:** Primary strategy uses Gemini text-embedding-004 (768-dim vectors). Falls back to local TF-IDF sparse vectors (512-dim) when API is unavailable.
- **Vector Store:** In-memory storage with source metadata for fast retrieval.
- **IR Search:** Query embedding compared via cosine similarity against all chunks; top-5 retrieved.
- **LLM Response:** Retrieved passages + user query sent to Gemini 2.0 Flash with strict grounding instructions to prevent hallucination.

3.3 Voice Navigation Architecture

The voice-first UX is implemented through a centralized command routing system:

- **Speech-to-Text (STT):** Android SpeechRecognizer with partial results for live feedback.
- **Command Parser:** VoiceCommandParser uses keyword matching to map speech into a sealed class hierarchy (25+ distinct commands).
- **Text-to-Speech (TTS):** AI responses are read aloud automatically (truncated at 400 chars for UX).
- **Command routing in MainActivity** dispatches to the correct ViewModel and navigates to the appropriate tab.

4. Features Implemented

4.1 Track 1: ESG Market Research Voice Navigator

Real-time ESG data visualization with voice-driven graph navigation and AI-powered analysis:

- GDP Growth Chart - World Bank API (NY.GDP.MKTP.KD.ZG) with line chart visualization.
- CO2 Emissions Chart - World Bank API (EN.ATM.CO2E.KT) with trend analysis.
- Agricultural Land Chart - World Bank API (AG.LND.AGRI.ZS) showing global land-use trends.
- AI-generated descriptions for each indicator via Gemini, with hardcoded fallback analysis.
- DOW Top 10 Stocks - Pie chart visualization with Gemini-powered stock analysis.
- Gas Price Time Series - BLS API (APU000074714) historical data from 1992-2026.
- Milk Price Time Series - BLS API (APU0000709112) historical consumer price data.
- Food Security Analysis - RAG-based Q&A from SOFI 2024/2025 reports with 7 preset query buttons.

4.2 Track 2: California DMV Knowledge Test Preparation

RAG-based compliance learning platform with voice interaction:

- Handbook Q&A - Voice or button-driven queries against CA Driver's Handbook content.
- 4 preset quick-query buttons: Signaling Signs, BAC Limits, Speed Limits, Right-of-Way.
- Free-form voice queries for any DMV topic (parking, headlights, DUI penalties, etc.).
- 15+ topic-specific fallback responses covering all major handbook sections.
- Practice Quiz - 10 multiple-choice questions with scoring, explanations, and progress tracking.
- Quiz covers: Speed Limits, DUI Laws, Parking, Traffic Signs, Signaling, Right-of-Way, Safe Driving.

4.3 Voice Navigation

25+ voice commands supported across all screens:

- Chart commands: 'Show GDP graph', 'Show CO2 chart', 'Show agricultural land'
- AI analysis: 'Describe CO2 emissions', 'Top 10 stocks of DOW'
- Food security: 'Major food security issues', 'Compare 2023 vs 2024'
- DMV queries: 'What are signaling signs', 'BAC limits', 'Speed limits'
- Navigation: 'Go to market research', 'Go to DMV', 'Go to ESG', 'Go home'

5. APIs and External Services

5.1 Google Gemini API

- Model: gemini-2.0-flash for chat completions (v1beta endpoint).
- Embeddings: text-embedding-004 for RAG vector generation.
- Used for: ESG indicator analysis, food security RAG, DMV handbook RAG, stock analysis, market research queries.
- Quota handling: Automatic retry with exponential backoff, quota exhaustion detection with cooldown timer.

5.2 World Bank API

- GDP Growth: NY.GDP.MKTP.KD.ZG - Annual percentage growth rate.
- CO2 Emissions: EN.ATM.CO2E.KT - Carbon dioxide emissions in kilotonnes.
- Agricultural Land: AG.LND.AGRI.ZS - Percentage of land area.
- 10-second timeout with static fallback chart data for offline resilience.

5.3 Bureau of Labor Statistics (BLS) API

- Gas Price: Series APU000074714 - US Regular All Formulations.
- Milk Price: Series APU0000709112 - Whole milk per gallon.
- Fetches in 20-year chunks to work within BLS API limits.

5.4 Android Speech APIs

- SpeechRecognizer for voice-to-text with partial results.
- TextToSpeech for reading AI responses aloud.

6. Project Structure

```

app/src/main/
  AndroidManifest.xml
  java/com/hackathon/voicenavigator/
    MainActivity.kt          - Entry point, navigation, voice routing
    VoiceNavigatorApp.kt    - Application class (PdfBox init)
  data/
    api/
      GeminiApiService.kt   - Gemini 2.0 Flash + embeddings API
      WorldBankApiService.kt - World Bank REST client
      BLSApiService.kt     - Bureau of Labor Statistics client
      OpenAIApiService.kt  - (Legacy, not currently used)
      RAGEngine.kt         - Full RAG pipeline: chunk/embed/search/LLM
    model/
      Models.kt            - All data models and enums
  viewModel/
    MarketResearchViewModel.kt - ESG charts, market data, caching
    ESGViewModel.kt         - Food security RAG + fallbacks
    DMVViewModel.kt         - DMV RAG + quiz + fallbacks
  voice/
    VoiceRecognitionManager.kt - STT/TTS + VoiceCommandParser
  ui/
    theme/ Color.kt, Theme.kt - SJSU brand colors, Material 3
    components/
      VoiceButton.kt        - Animated voice FAB
      Charts.kt             - LineChart + PieChart (Canvas)
      BottomNavBar.kt       - Navigation bar + TopBar
    screens/
      MarketResearchScreen.kt - API tab (GDP/CO2/Agri/BLS)
      ESGDashboardScreen.kt   - ESG tab (Food Security RAG)
      DMVScreen.kt           - DMV tab (Handbook RAG + Quiz)

```

7. Key Design Decisions

- **Voice-First UX:** Every screen has a prominent voice button; all major actions are voice-accessible. TTS reads back AI responses for a touchless experience.
- **Dual Embedding Strategy:** Gemini neural embeddings (768-dim) as primary, with automatic fallback to local TF-IDF (512-dim). Ensures RAG works even without API quota.
- **Comprehensive Fallbacks:** Every API call and AI query has hardcoded fallback responses. The app is fully functional offline for demos.
- **Response Caching:** In-memory caching in each ViewModel prevents redundant API calls and ensures instant responses on repeated queries.
- **Custom Canvas Charts:** Zero external charting library dependencies. LineChart and PieChart rendered directly on Compose Canvas.
- **Material 3 Theming:** Modern Android design with SJSU brand colors (blue/gold) and consistent card-based UI.
- **Inline Knowledge Base:** ESG and DMV content embedded as string constants for reliable, instant access without PDF parsing at runtime.

8. Dependencies

Library	Purpose
androidx.compose.*	Jetpack Compose UI framework
androidx.navigation	Compose navigation
androidx.lifecycle	ViewModel + lifecycle management
com.squareup.retrofit2	REST API client (BLS)
com.squareup.okhttp3	HTTP client + logging
com.google.code.gson	JSON serialization
kotlinx.coroutines	Asynchronous programming
com.tom-roush:pdfbox-android	PDF text extraction
io.coil-kt:coil-compose	Image loading (available)
androidx.webkit	WebView support

9. Setup and Running

- Prerequisites: Android Studio Hedgehog+, JDK 17, Android SDK 34, physical device or emulator (API 26+).
- Clone/import the project into Android Studio.
- Set Gemini API key in local.properties: GEMINI_API_KEY=your-key-here
- (Optional) Place PDF files in app/src/main/assets/ for full RAG pipeline.
- (Optional) Set BLS API key for higher rate limits.
- Sync Gradle, build, and run on device. Grant microphone permission when prompted.

10. References

- C. Vuppapapati, Machine Learning and AI for Agricultural Economics. Springer, 2021.
- C. Vuppapapati, Building Next-Gen ESG Platforms with IoT and AI for SDGs. CRC Press, 2026.
- World Bank Open Data - <https://data.worldbank.org>
- FAO/WHO/UNICEF - The State of Food Security and Nutrition in the World 2024 & 2025.
- California DMV - Driver's Handbook (<https://www.dmv.ca.gov>)
- Bureau of Labor Statistics - <https://api.bls.gov>
- Google Gemini API - <https://ai.google.dev>